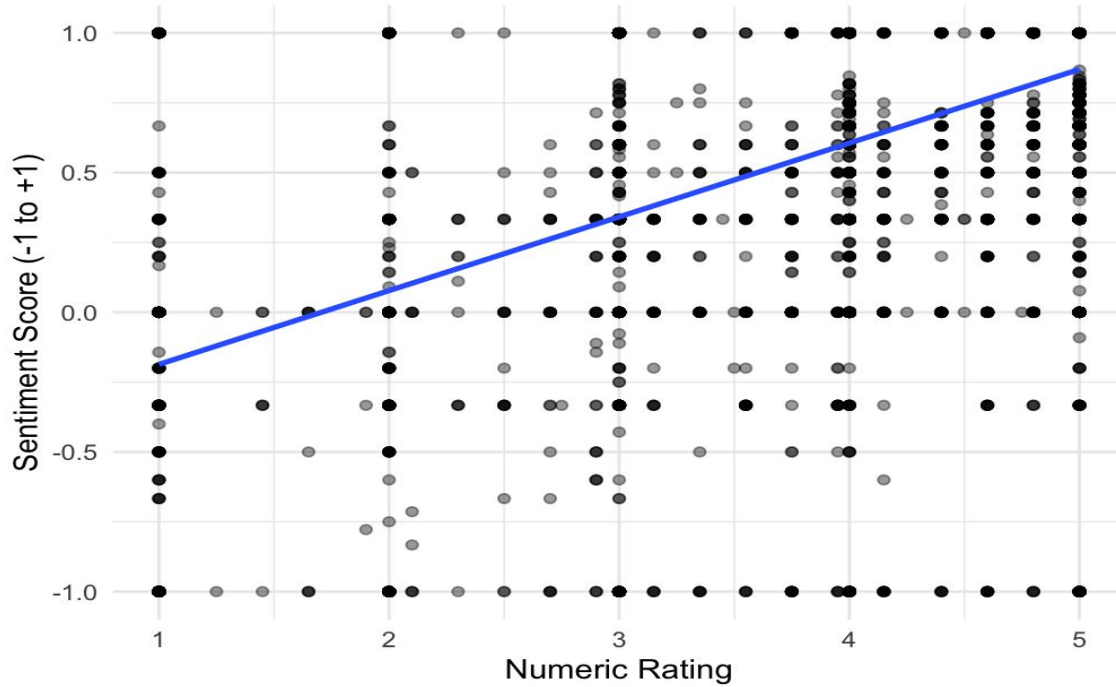


Hotel Reviews Assignment

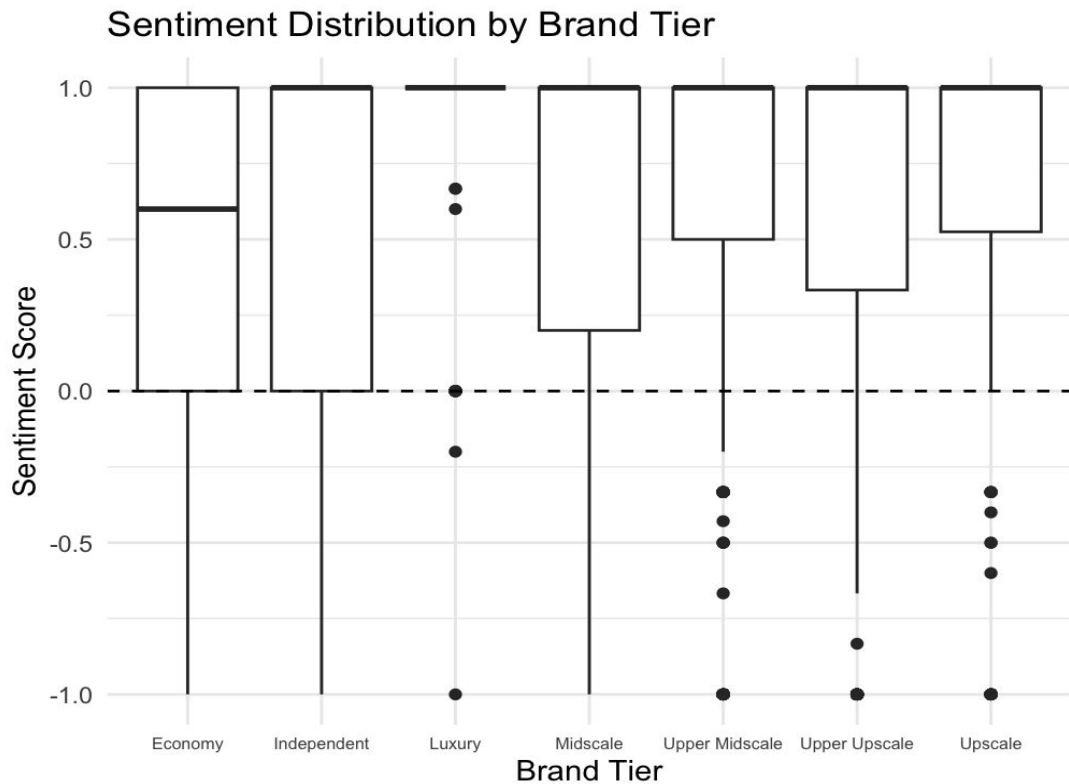
Gregory Beard

Visualization 1

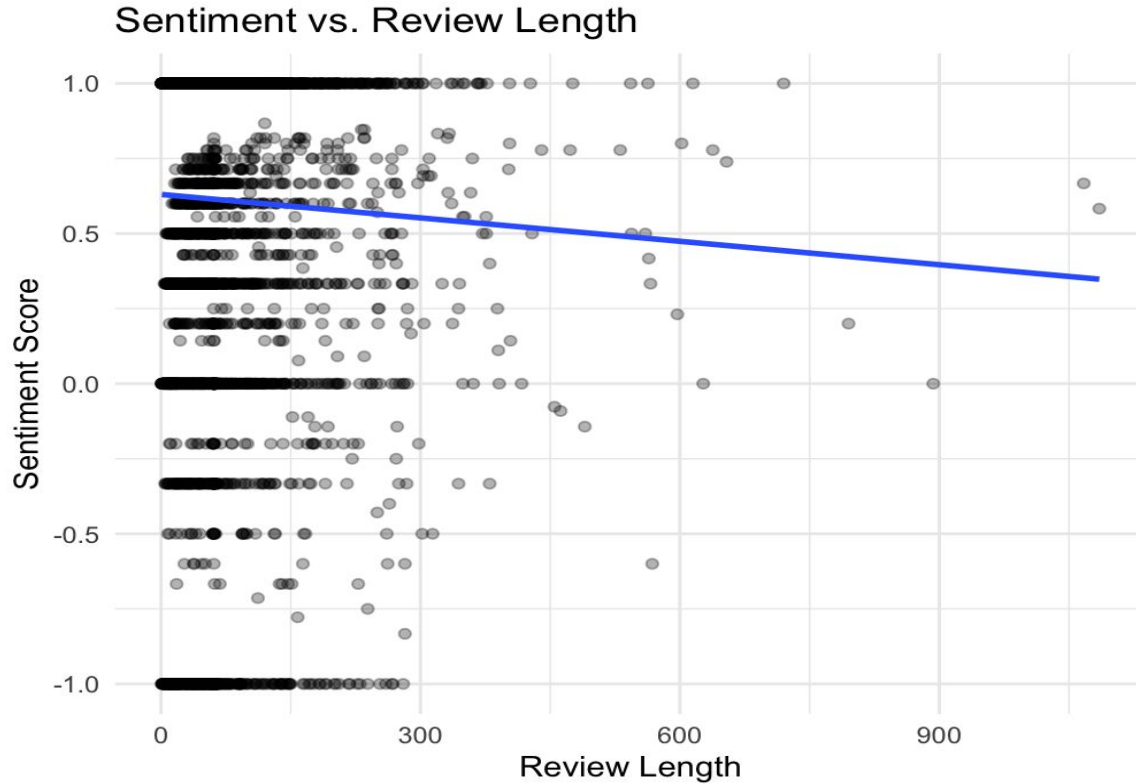
Sentiment Score vs. Numeric Rating



Visualization 2



Visualization 3



Incongruity 1

Sentiment Score: -1

Numeric Rating: 5

Explanation: It seems like the only problem the customer had was with the thermostat in the room. The mismatch may exist because the review text only referenced the problem the customer had with the air conditioning and thermostat. One would assume everything else was up to par because the review score was still a 5.

Incongruity 2

Sentiment Score: -1

Numeric Rating: 4.15

Explanation: In this customer's review, the only complaint was the bed being uncomfortable. Similarly to the previous incongruity, one can be led to believe the mismatched ratings stem from the fact that the review text was mostly negative, which is the only thing the sentiment score accounts for.

Incongruity 3

Sentiment Score: 1

Numeric Rating: 2

Explanation: This mismatch likely occurs because of the way the customer words their review. There is a bit of a mixed tone. The first sentence talks negatively about the hotel, but uses words that don't normally get used in hotel reviews such as "barebones". The review goes on to talk about the cleanliness of the hotel, along with the friendliness of the staff, which likely is what led to the computer generated sentiment score of 1.

Interpretation

Sentiment analysis does not meaningfully align with numeric ratings in this dataset. As seen in Visualization 1, there may be a very small correlation that exist, however, I do not think it is strong enough to say there is meaning alignment between the two variables. Although a good chunk of the positive ratings are associated with higher sentimental scores, there are also a chunk of positive ratings that have low sentimental scores (and vice versa). Some of these are due to sarcasm, mixing tones, emphasis on one critical issue, or flat out misinterpretation of ambiguous wording. Overall, I think there is a small, but weak relationship. There would definitely need to be additional research done before saying there is meaningfully and strong alignment in this dataset. This highlights the limitations of automated sentiment scoring because reviews can be ambiguous or sarcastic. As a result, the sentiment score can be misleading and skew datasets. In order to make it more effective, there would need to be a method put into place that allows for context to be accounted for in automated sentiment scoring. With that being said, it is safe to say that cost proxy appears to influence tone. As seen in Visualization 2, The distribution of sentiment scores shifts upward as brand tier increases, suggesting that higher-cost tiers are associated with more positive tone.

Appendix

Prompt Used:

“Looking at the reviews.text column, perform a sentiment analysis. Assign a sentiment score between -1 (very negative) and 1 (very positive) based on the review. Return only the numeric score. You may modify wording, but the scale must remain consistent”.